

❖ Operators in Python

- Operators are special symbols or keywords that perform operations on one or more operands (values or variables) to produce a result.

- operand1 operator operand2

❖ Types of Operators in Python

- 1) Assignment Operators
- 2) Arithmetic Operators
- 3) Comparison Operators
- 4) Logical Operators
- 5) Identity Operators
- 6) Membership Operators

1. Assignment Operators

- Assignment Operators are used to assign values to variables. They can also perform an operation and assign the result back to the same variable.

Operator	Name	Example	Equivalent To
=	Assignment	x = 10	Assign 10 to x
+=	Add and Assign	x += 5	x = x + 5
-=	Subtract and Assign	x -= 5	x = x - 5
*=	Multiply and Assign	x *= 5	x = x * 5
/=	Divide and Assign	x /= 5	x = x / 5
//=	Floor Divide and Assign	x //= 5	x = x // 5
%=	Modulus and Assign	x %= 5	x = x % 5
**=	Exponent and Assign	x **= 2	x = x ** 2

2. Arithmetic Operators

- Arithmetic operators are symbols used to perform basic mathematical operations like addition, subtraction, multiplication, division, etc.

Operator	Name	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5
//	Floor Division	10 // 3
%	Modulus	10 % 3
**	Exponent (Power)	2 ** 3

3. Comparison Operators

- Comparison Operators are used to compare two values or variables. The result of a comparison is always either True or False.

Operator	Name	Meaning
==	Equal To	Checks whether two values are equal
!=	Not Equal To	Checks whether two values are different
>	Greater Than	Checks whether left value is greater
<	Less Than	Checks whether left value is smaller
>=	Greater Than or Equal To	Checks greater or equal
<=	Less Than or Equal To	Checks smaller or equal

4. Logical Operators

- Logical operators are used when we need to evaluate more than one condition.

Operator	Name	Description	Example
and	Logical AND	Returns True if both conditions are True	(a > 5 and b < 10)
or	Logical OR	Returns True if at least one condition is True	(a > 5 or b < 10)
not	Logical NOT	Reverses the Boolean value (True → False, False → True)	not(a > 5)

5. Identity Operators

- Identity operators compare the memory location (identity) of two objects, not just their values.

Operator	Name	Description	Example
is	Identity Operator	Returns True if both variables refer to the same object	a is b
is not	Negative Identity Operator	Returns True if both variables refer to different objects	a is not b

6. Membership Operators

- Membership operators check whether a value is present or absent in a collection (string, list, tuple, set, or dictionary).

Operator	Name	Description	Example
in	Membership Operator	Returns True if the element is present in the sequence	"a" in "apple"
not in	Negative Membership Operator	Returns True if the element is not present in the sequence	"z" not in "apple"